

Module 3: Basic Statistics — Summary

What this module is about. Measures of center tell us where a data set is located, while measures of spread tell us how tightly or loosely the values cluster. A good description usually needs both. The mean and standard deviation use every observation and are sensitive to extreme values. The median is based on order and is often more resistant to an outlier.

For data $x_1, \dots, x_n$:

$$\bar{x} = \frac{x_1 + \dots + x_n}{n}, \quad \text{range} = \max - \min.$$

The **median** is the middle ordered value (or the average of the two middle values). The **mode** is the most frequent value.

Throughout this module, use the Chapter 3 standard-deviation convention, which divides by n :

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}.$$

Example 1: Center and an outlier

A store records complaints on five days:

2, 3, 2, 4, 9.

The mean is

$$\bar{x} = \frac{2 + 3 + 2 + 4 + 9}{5} = 4,$$

while the median is 3 and the mode is 2. The value 9 pulls the mean upward. If the unusually high day is removed, the remaining data are 2, 2, 3, 4, with mean 2.75 and median 2.5. The center changes, but the mean changes more noticeably because it uses the magnitude of every value.

Example 2: Standard deviation measures consistency

For the data 3, 5, 5, 7, the mean is 5. The deviations and squared deviations are

$$-2, 0, 0, 2 \quad \implies \quad 4, 0, 0, 4.$$

So

$$s = \sqrt{\frac{4 + 0 + 0 + 4}{4}} = \sqrt{2} \approx 1.41.$$

Now compare two filling machines with the same mean of 250 mL:

$$A : 248, 250, 252, 250, \quad B : 244, 250, 256, 250.$$

Machine A has the smaller spread, so it is the more consistent machine even though both machines have the same mean.

1. Compute the mean $\bar{x}$.
2. Find each deviation $x_i - \bar{x}$.
3. Square the deviations.
4. Add the squared deviations.
5. Divide by n .
6. Take the square root.

The units of standard deviation are the same as the units of the original data.

By the end of this module, you should be able to: compute mean, median, mode, range, and the Chapter 3 standard deviation; recover a missing value from a known mean; compare consistency using spread; and explain why an outlier can affect the mean more strongly than the median.

Module 3: Basic Statistics — Prequiz

Open-note. Show your mathematical work. A calculation and a clearly stated final answer are enough unless a sentence is specifically requested.

Name:

Date:

Standard deviation for this module

$$\bar{x} = \frac{\sum x_i}{n}, \quad s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}.$$

1. A small store records the number of customer complaints received on five consecutive days:

3, 4, 3, 5, 10.

Find the mean, median, mode, and range of the five values.

2. The value 10 in Problem 1 came from an unusually difficult day. Remove it and use the remaining data 3, 3, 4, 5.

- (a) Find the new mean and median.
- (b) Which measure of center changed more when the outlier was removed: the mean or the median?

3. Five students have a mean test score of 82. Four of the scores are 74, 80, 86, 83.

- (a) What total number of points must the five scores have altogether?
- (b) What is the missing fifth score?

4. Using the Chapter 3 convention shown above, compute the standard deviation for the data

4, 6, 6, 8.

Show the mean, the squared deviations, their sum, and the final square root.

5. Two machines each average 250 mL per bottle.

$$A : 247, 250, 253, 250, \quad B : 240, 250, 260, 250.$$

Without doing a full standard-deviation table, use the ranges to decide which machine is more consistent. Compute both ranges and state your conclusion.

Module 3: Basic Statistics — Matching Quiz

These problems use the same skills as the prequiz, with new values and contexts. Show enough work to make your reasoning visible.

Name:

Date:

1. A café records the number of orders received during five ten-minute periods:

6, 7, 6, 8, 13.

Find the mean, median, mode, and range.

2. Remove the unusually large value 13 from Problem 1. Find the new mean and median. Which of the two measures changes more?
3. Six students have a mean score of 75. Five scores are 68, 72, 76, 80, 74. Find the missing sixth score.
4. For the data 2, 4, 4, 6, the mean is 4. Using the Chapter 3 convention, compute the standard deviation by finding and averaging the squared deviations, then taking a square root.
5. Set A is 49, 50, 50, 51, 50. Set B is 30, 40, 50, 60, 70. Both have mean 50. Which set is more consistent, and what feature of the data supports your answer?